

My research is in model theory, a branch of mathematical logic, and constraint satisfaction problems (CSPs), a topic in theoretical computer science. In model theory, I work on interactions between model theory, combinatorics, and the study of probabilistic symmetries. On CSPs, I study computational problems which are naturally represented on well-behaved infinite structures.

Positions

- 2026-present **Senior Research Associate**, Department of Computer Science. University of Oxford.
Funded by NAASP, UKRI ERC Guarantee Grant (previously ERC Consolidator Grant).
Research Member of the Senior Common Room of Worcester College.
- 2023-2026 **Postdoc**, Institut für Diskrete Mathematik & Geometrie. TU Wien. Vienna.
Funded by POCOCOP, ERC Synergy Grant No. 101071674.

Education

- 2019-2023 **PhD in Pure Mathematics**, Imperial College London.
Admin Roth Scholarship. Supervisors: Prof. David Evans, Dr. Charlotte Kestner.
Thesis: Measures and amalgamation properties in ω -categorical structures.
- 2018 **Summer Research Project**, University of Oxford.
Topic: The Separation of Right Angled Artin Groups (geometric group theory);
with Dr. Richard Wade.
- 2015-2019 **Master of Mathematics and Philosophy**, University of Oxford, Worcester College.
Parts A & B: First Class. Part C: First Class. (Parts A & B correspond to my undergraduate studies).
Dissertation in mathematics: Categoricity for totally transcendental theories;
with Prof. Ehud Hrushovski.
Thesis in philosophy: The metaontology of thin objects; with Prof. James Studd.

Articles

- 2025 **Binary symmetries of tractable non-rigid structures**. Joint with Michael Pinsker. Fortieth Annual ACM/IEEE Symposium on Logic in Computer Science (LICS). This is a conference version of "Minimal operations over permutation groups".
- 2025 **Invariant Keisler measures for ω -categorical structures**. The Journal of Symbolic Logic. Volume 90 , Issue 2. pp. 791-807.
- 2025 **On the non-measurability of ω -categorical Hrushovski constructions**. Archive for Mathematical Logic. Volume 64, pp. 351–386.

Preprints

- 2026 **All mixed identities are singular in groups with no algebraicity**. Joint with Michael Pinsker. ArXiv:
- 2026 **Minimal and intrinsic topologies on monoids of elementary embeddings**. Joint with J de la Nuez Gonzales, Zaniar Ghadernezhad, and Michael Pinsker. ArXiv: 2603.28419.
- 2025 **Taking model complete cores**. Joint with Manuel Bodirsky and Bertalan Bodor. ArXiv: 2512.21278. Submitted.
- 2025 **A guide to topological reconstruction on endomorphism monoids and polymorphism clones**. Joint with Michael Pinsker. ArXiv:2512.01086. In revision at Springer Series: Outstanding Contributions to Logic, volume in honour of Mai Gehrke. Reviewer recommends for publication.
- 2024 **Minimal operations over permutation groups**. Joint with Michael Pinsker. ArXiv:2410.22060. The conference version of this paper, "Binary symmetries of tractable non-rigid structures" is in the proceedings of LICS 2025. In revision at Journal of Algebra. Reviewer recommends for publication.
- 2024 **When invariance implies exchangeability (and applications to invariant Keisler measures)**. Joint with Samuel Braunfeld and Colin Jahel. ArXiv:2408.08370. In revision at Annales Henri Lebesgue. Reviewer recommends for publication.

Selected talks and posters (★ :=invited talks in person)

- ★ Jun 2026 **Mixed identities and Neumann's Lemma.** LyMoTs meeting. University of Leeds.
- ★ May 2026 **Minimal and intrinsic topologies on monoids of elementary embeddings.** Heilbronn Semigroups Seminar. University of Manchester.
- ★ Apr 2026 **Mixed identities and Neumann's Lemma.** Back & forth lectures. University of Turin. Outreach talk for undergraduate students.
- ★ Jan 2026 **Taking model-complete cores.** London Logic Seminar. Imperial College London.
- Dec 2025 **Tutorial: on the topological reconstruction of endomorphism monoids of ω -categorical structures.** Oberwolfach Workshop: Homogeneous Structures: Model Theory meets Universal Algebra. Extended abstract published in the meeting report.
- ★ Nov 2025 **On topological reconstruction for monoids of elementary embeddings.** Logic Seminar. University of Leeds.
- ★ Sep 2025 **Minimal operations over permutation groups.** 63rd Summer School on General Algebra and Ordered Sets (SSAOS). Blansko, Czech Republic.
- Jul 2025 **When invariance implies exchangeability.** Logic Colloquium 2025. TU Wien. An abstract of this talk will appear in the meeting report published by the Bulletin of Symbolic Logic.
- Jun 2025 **Binary Symmetries of tractable non-rigid structures.** 40th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS). Singapore. Paper selected through peer-review.
- Nov 2024 **Minimal operations over permutation groups.** PALS, Panglobal Algebra and Logic Seminar. University of Colorado.
- Oct 2024 **When invariance implies exchangeability.** Peking University Model Theory seminar.
- Jul 2024 **Exchangeability of consistent random expansions.** Midsummer Combinatorial Workshop XXIX, Prague.
- May 2024 **Minimal operations over permutation groups.** AAA105, Prague.
- ★ Mar 2024 **Minimal operations over permutation groups.** Algebra seminar. Korean Institute of Advanced Studies.
- ★ Mar 2024 **When measures don't care about structure (and when they do).** Logic seminar. Yonsei University.
- ★ Sep 2023 **Measures in homogeneous 3-hypergraphs.** Model Theory Workshop. Wroclaw.
- Jun 2023 **Measures in ternary homogeneous structures.** SEEMOD. University of Oxford.
- Jan 2023 **Invariant Keisler measures in simple ω -categorical structures.** Logic seminar. Carnegie Mellon University.
- ★ Nov 2022 **Universally measure zero non-forking formulas in simple omega-categorical Hrushovski constructions.** 7th French-Kazakh Colloquium in Model Theory. Institut Camille Jordan. Lyon.
- Jul-Aug 2022 Poster: **Invariant Keisler Measures in ω -categorical Hrushovski Constructions.** UNIMOD 2022. I also contributed by typing the exercises for the problem classes.
- Mar 2022 **Non-measurability of omega-categorical Hrushovski constructions.** European Conference on Interdisciplinary Model Theory - ECIMT. Münster.
- ★ Dec 2021 **Non-MS-measurability of omega-categorical Hrushovski constructions.** South and East of England Model Theory Network Conference (SEEMOD). Queen Mary University of London.

Long research visits

- Mar 2024 **Two weeks visit at the Korean Institute of Advanced Studies (KIAS).** Visiting J de la Nuez Gonzales. Funded by the KIAS.
- Nov 2022 **One month visit to the Institut Camille Jordan (ICJ),** Université Claude Bernard, Lyon 1, France. Funded by the ICJ.

Organisation of meetings and conferences

- 2026 **Arbeitstagung Allgemeine Algebra (AAA) 108 - 108th Workshop on General Algebra.** TU Wien. Organizer jointly with Michael Pinsker and Mike Behrisch. Secured 4800€ grant from the Vienna Meeting Fund.
- 2024, 2025 **POCOCOP meeting.** TU Wien. Organiser. Room, hotel, and restaurant bookings. Schedule.

- 2025 **Shelah's 80th birthday conference & VOrST (Vienna Oracle of Set Theory)**. TU Wien. Registration and coffee breaks.
- 2022 **SEEMOD**. Imperial College London.

Organisation of reading groups involving multiple universities

- 2023-2025 **Finite Structures with Few Types**, by Cherlin and Hrushovski. Jointly with Aris Papadopoulos. TU Wien, U. Leeds, U. Notre Dame, Imperial, and U. Maryland.
- 2023-2024 **CSP reading group**. Organiser. TU Wien, TU Dresden, Charles University (Prague), U. Leeds, and U. Cambridge. Website with recordings: cspreadinggroup.github.io.
- 2023 **On NSOP₂ theories**, by Mutchnik. Organiser. Imperial and Institut Camille Jordan.

Awards

- 2026 **Marie Skłodowska-Curie Actions (MSCA) Seal of Excellence**. For the project "Partial Exchangeability, Regularity, and Higher Arity Model Theory". University of Cambridge.
- 2023 **Runner-up prize for the Mathematics PhD Symposium poster competition**. Imperial College London.
- 2022 **Travel Grant from the Association for Symbolic Logic (ASL)** for Unimod 2022.
- 2017 **Gibbs Prize for FHS Mathematics and Philosophy Part A**. Best performance in Math.
- 2016 **Scholarship, Worcester College**. For performance in first year exams.

Teaching

- 2024-2025 **Model theory II: Stability**. TU Wien. Masters' course. Designing and teaching the course. 16 lectures+problem classes & oral exam. Lecture notes and recordings available on my website.
- 2023-2025 **Supervising students** at TU Wien:
- bachelor thesis on "Automorphism groups and Ramsey properties of sparse graphs" by Evans, Hubička, and Nešetřil;
 - bachelor mini-thesis on "The CSP Dichotomy, the Axiom of Choice, and Cyclic Polymorphisms" by Kátay, Tóth, and Vidnyánszky;
 - visiting student (from ISEER Bhopal) bachelor project on Hall's universal group.
- 2024 **Model theory**. TU Wien. Masters' course. Problem classes, problem sheets, and two lectures.
- 2022 **Associate Fellowship of the Higher Education Academy**. Approved through the Imperial College London STAR Framework.
- 2019-2023 **Graduate Teaching Assistant** at Imperial College London: Introduction to University Mathematics (2020, 2022); Introduction to University Mathematics re-sit exam preparation (2021, 2023); Linear Algebra and Groups (2019-2022); Latex workshops for first year projects (2020-2022); Mathematics Pre-Arrival Course (2021-2022); Commutative Algebra (2020).
- 2020-2022 **Senior Graduate Teaching Assistant**: Analysis II (metric spaces and topology).
- 2020-2021 **Personal tutoring** in mathematics and philosophy. Grosvenor Tutors.

Professional development courses

- 2019-2023 Graduate School and Department of Mathematics of Imperial College London: Teaching: Maths GTA training; Intro to Assessment and Feedback for Learning; Intro to Learning and Teaching; Performative Aspects of Teaching; Microteaching; GTA Retreat for Associate Fellowship to the Higher Education Academy. Management Skills: Time Management for your Doctorate.

Other skills and interests

- Languages:** Fluent: Italian, Spanish, English. Elementary (A1): German, French.
- Software:** **For mathematics:** $\text{\LaTeX}$. Lean, GAP, and MAGMA.
Graphics: Adobe Creative Cloud: Illustrator; Photoshop; Indesign. Inkscape.
- Philosophy:** Early analytic philosophy (Frege, Russell, Wittgenstein, and Ramsey). Philosophy of logic, mathematics, and quantum physics. Ethics (Anscombe, Murdoch, Foote, and Midgley).
- Origami:** Advanced practitioner and creator. Interested in uses of origami for mathematics education.